

# DEEP LEARNING IMPORTANT QUESTIONS 2026-2027(IV-I)

| UNIT I  |                                                                                                                                                                                                                                                                                                       |    |     |       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-------|
| 1       | Describe the McCulloch-Pitts neuron model. Explain its assumptions, architecture, mathematical model, advantages and limitations.                                                                                                                                                                     | 5M | CO1 | BTL2  |
| 2       | Using Threshold Logic, design McCulloch-Pitts (MCP) neurons to implement the AND, OR, and NOT logical gates. Specify the input and threshold for each neuron and verify the output for all possible input combinations. Draw a neat diagram of multi-layer perceptron model to solve the XOR problem. | 5M | CO1 | BTL3  |
| 3       | Explain the architecture of the Perceptron and derive the Perceptron Learning Algorithm, illustrating its working with a suitable example.                                                                                                                                                            | 5M | CO1 | BTL2  |
| 4       | Given a perceptron has initial weights $w_1 = 0.2$ , $w_2 = -0.3$ , bias 0.1, learning rate 0.2. For input (1,1) with target output 1, perform one iteration of the Perceptron Learning Algorithm and calculate the updated weights and bias.                                                         | 5M | CO1 | BTL3  |
| 5       | Explain the Representation Power of Multi-Layer Perceptron (MLPs).                                                                                                                                                                                                                                    | 5M | CO1 | BTL-4 |
|         |                                                                                                                                                                                                                                                                                                       |    |     |       |
|         |                                                                                                                                                                                                                                                                                                       |    |     |       |
| UNIT II |                                                                                                                                                                                                                                                                                                       |    |     |       |
| 6       | Compute one complete iteration of the Backpropagation algorithm using the feed-forward neural network given below..                                                                                                                                                                                   | 5M | CO2 | BTL3  |

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| 7                                             | <p>Compute the covariance matrix and first principal component for a given 2-D dataset.</p> <table border="1"> <thead> <tr> <th>Sample</th> <th>(X1)</th> <th>(X2)</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>1</td> <td>2</td> </tr> <tr> <td>2</td> <td>2</td> <td>3</td> </tr> <tr> <td></td> <td></td> <td></td> </tr> <tr> <td>3</td> <td>3</td> <td>5</td> </tr> </tbody> </table> | Sample | (X1) | (X2) | 1 | 1 | 2 | 2 | 2 | 3 |  |  |  | 3 | 3 | 5 | 5M | CO1 | BTL4 |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|------|---|---|---|---|---|---|--|--|--|---|---|---|----|-----|------|
| Sample                                        | (X1)                                                                                                                                                                                                                                                                                                                                                                                          | (X2)   |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 1                                             | 1                                                                                                                                                                                                                                                                                                                                                                                             | 2      |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 2                                             | 2                                                                                                                                                                                                                                                                                                                                                                                             | 3      |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                               |        |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 3                                             | 3                                                                                                                                                                                                                                                                                                                                                                                             | 5      |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 8                                             | <p>Compute the Singular Value Decomposition of <math>A = \begin{bmatrix} 3 &amp; 1 &amp; 1 \\ -1 &amp; 3 &amp; 1 \end{bmatrix}</math> and reconstruct the matrix.</p>                                                                                                                                                                                                                         | 5M     | CO1  | BTL3 |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 9                                             | <p>Using Momentum GD (<math>\beta=0.9</math>), perform two iterations for <math>f(x)=x^2</math> starting from <math>x=2</math>.</p>                                                                                                                                                                                                                                                           | 5M     | CO1  | BTL3 |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 10                                            | <p>A neuron has <math>x=[1,2]</math>, <math>w=[0.4,0.6]</math>, bias=0.2, <math>\eta=0.1</math>. Perform one iteration of gradient descent and compute updated weights.</p>                                                                                                                                                                                                                   |        | CO1  | BTL3 |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| <h2 style="text-align: center;">UNIT III</h2> |                                                                                                                                                                                                                                                                                                                                                                                               |        |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 11                                            | <p>Establish the relationship between a linear autoencoder and Principal Component Analysis (PCA) by explaining the conditions under which they are equivalent and how the encoder weights correspond to the principal components.</p>                                                                                                                                                        | 5M     | CO2  | BTL3 |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
|                                               |                                                                                                                                                                                                                                                                                                                                                                                               |        |      |      |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 12                                            | <p>Explain the working principle of a Denoising Autoencoder by describing how noisy inputs are generated and used to reconstruct the corresponding clean inputs, and formulate its training objective with a suitable example.</p>                                                                                                                                                            | 5M     | CO2  | BTL2 |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |
| 13                                            | <p>Explain the working principle of a Contractive Autoencoder by describing the motivation for contractive regularization and how the Jacobian-based penalty makes the learned</p>                                                                                                                                                                                                            | 5M     | CO2  | BTL2 |   |   |   |   |   |   |  |  |  |   |   |   |    |     |      |

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|    |                                                                                                                                                                                                                                                                                           |    |     |       |
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|    | representation less sensitive to small variations in the input, with a suitable example                                                                                                                                                                                                   |    |     |       |
| 14 | For the following input and reconstructed output, $x = [1,2,3]^T$ and $\hat{x} = [0.8,2,2.5]^T$ calculate the Mean Squared Reconstruction Error. Explain how this error is used for training an Autoencoder.                                                                              | 5M | CO2 | BTL3  |
| 15 | Given an Autoencoder with input $x = [1,2,3]^T$ , encoder $h = g(Wx + b)$ and decoder, $\hat{x} = f(W'h + c)$ , where the weights, biases, and activation functions are specified, calculate the hidden representation and reconstructed output. Also determine the reconstruction error. | 5M | CO2 | BTL-3 |